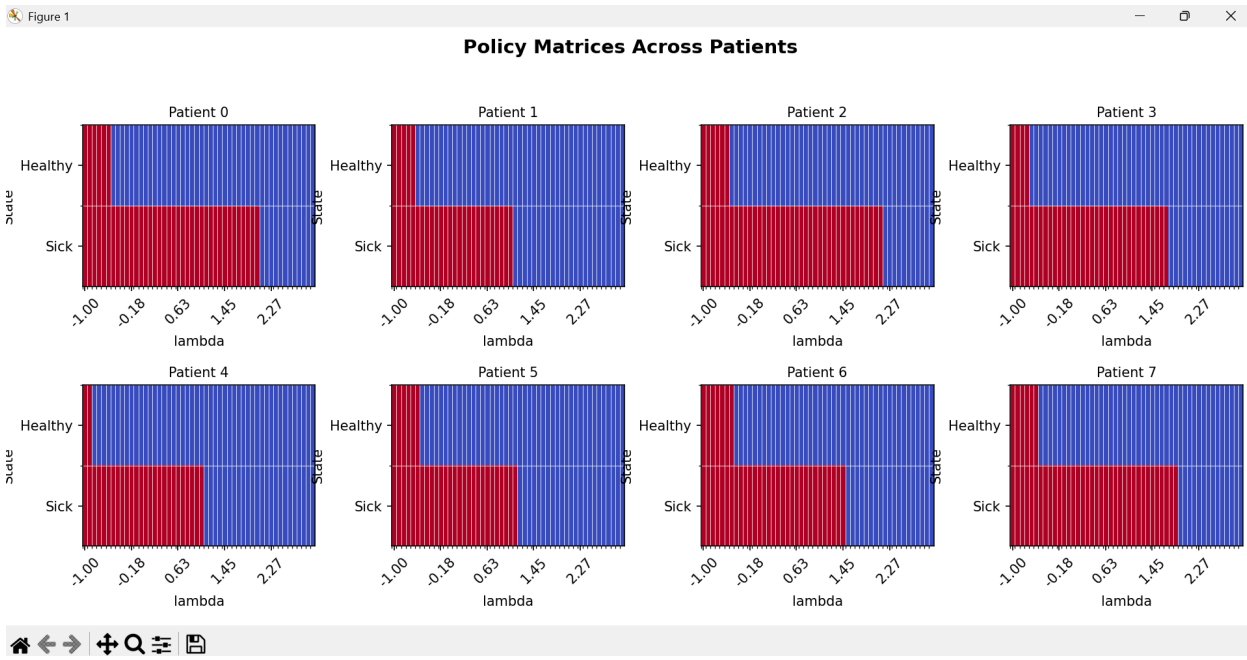
Setup -> over **200 days, k=10, N=100, beta=0.95**

Sample Whittle Policy Figure:
(First 8 out of 100 synthetically generated patients)



(Red -> optimal to schedule; Blue -> optimal to wait)

1 Simulation run

```
SIMULATION SUMMARY
Total Days: 200
Total Undiscounted Reward: 13258.50
Total Discounted Reward: 1308.50
Average Reward per Day: 66.29

Fairness Check (Longest Wait >= 50):
total patients with fairness violation: 65 out of 100
Patients with fairness violation:
```

65/100 patients waited more than 50 days for their next checkup.
60/100 patients never got scheduled.

Compared over 50 simulations

```
============================================================
RUNNING COMPARISON OVER MULTIPLE SIMULATIONS
============================================================
Sim 10/50 | Reward: 1343.91 | Violations: 65
Sim 20/50 | Reward: 1347.83 | Violations: 65
Sim 30/50 | Reward: 1328.01 | Violations: 65
Sim 40/50 | Reward: 1348.67 | Violations: 65
Sim 50/50 | Reward: 1286.02 | Violations: 65

Simulations: 50
Days per simulation: 200
Average Discounted Reward: 1318.22
Std Dev: 21.79
Min: 1280.59
Max: 1378.91

Fairness (wait >= 50):
  Avg violations per sim: 64.98
  Sims with any violation: 50/50
  Avg never scheduled per sim: 60.10
```

Running comparisons with **beta=0.85**

```
============================================================
RUNNING COMPARISON OVER MULTIPLE SIMULATIONS
============================================================
Sim 10/50 | Reward: 431.18 | Violations: 64
Sim 20/50 | Reward: 413.18 | Violations: 64
Sim 30/50 | Reward: 425.15 | Violations: 65
Sim 40/50 | Reward: 418.44 | Violations: 63
Sim 50/50 | Reward: 412.56 | Violations: 64

Simulations: 50
Days per simulation: 200
Average Discounted Reward: 426.08
Std Dev: 11.30
Min: 403.47
Max: 452.36

Fairness (wait >= 50):
  Avg violations per sim: 64.18
  Sims with any violation: 50/50
  Avg never scheduled per sim: 59.64
```

On average, there were around **60 violations of the fairness limit/100 patients.**

Now,
Put a **hard constraint** that **if wait time is over 50**, the patient must get scheduled.

With beta=0.85

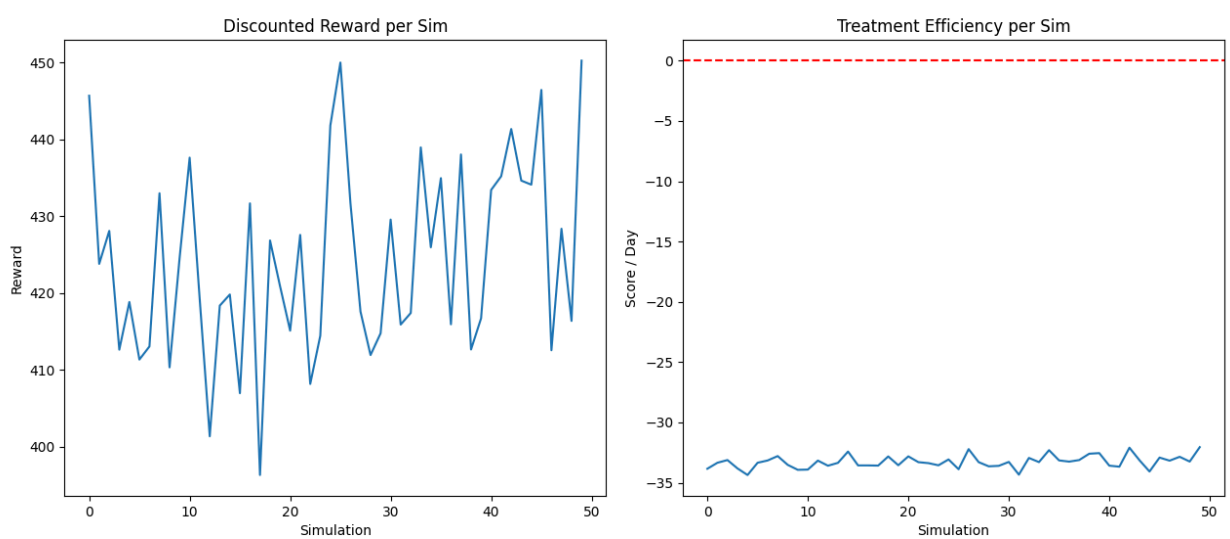
```
Simulations: 50
Days per simulation: 200
Average Discounted Reward: 421.97
Std Dev: 11.79
Min: 398.83
Max: 447.23

Fairness (wait >= 50):
  Avg violations per sim: 64.54
  Sims with any violation: 50/50
  Avg never scheduled per sim: 0.00
```

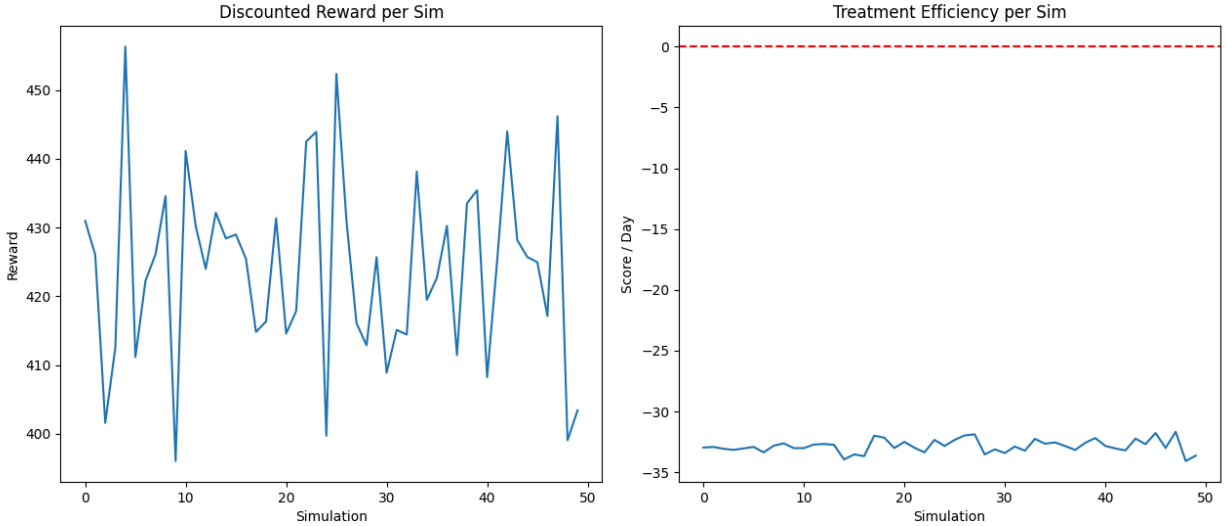
The average discounted reward went down by about 5 points.
But everyone got scheduled.

Comparing Results of Hard Constraint v/s Normal Whittle RMAB
Beta =0.85

1. Hard Constraint



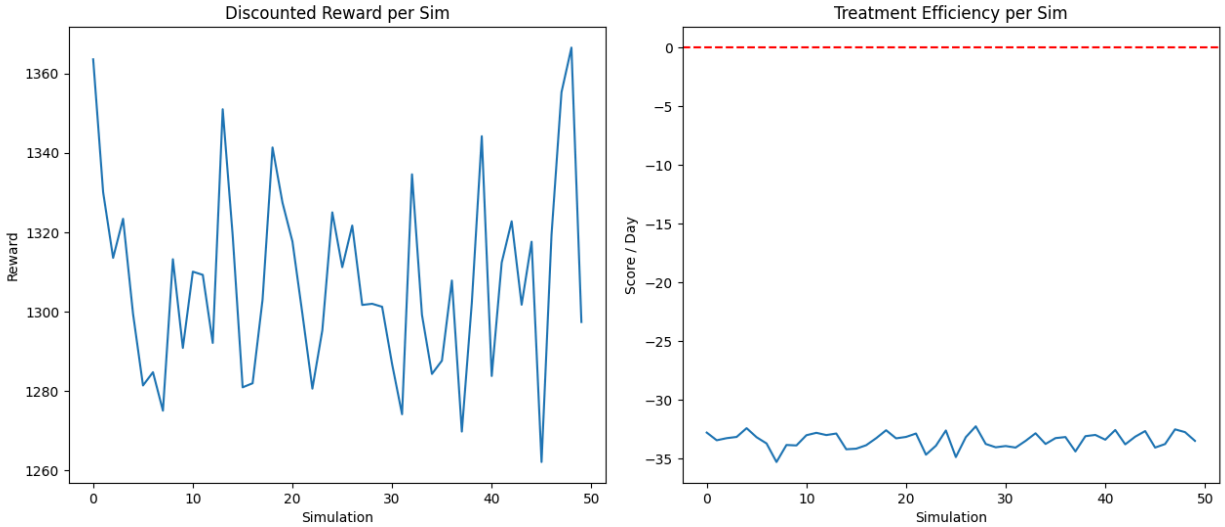
2. Normal RMAB



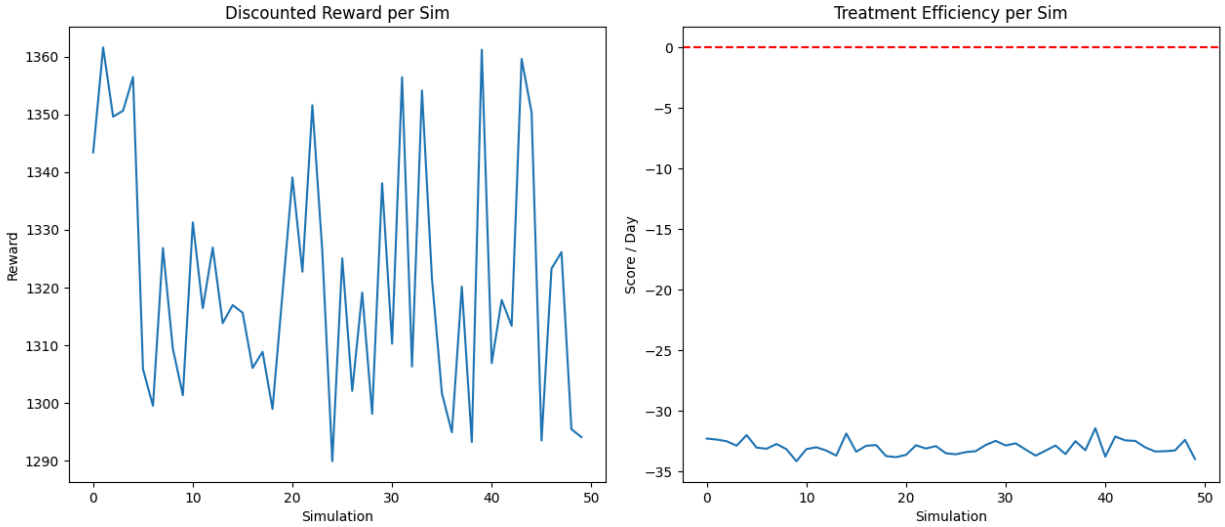
Treatment efficiency (the average number of actually sick patients scheduled - number of actually sick patients not scheduled) is more or less the same.
However, the discounted reward is better for normal whittle.

Similar trends **for beta=0.95**

1. Hard constraints



2. Normal RMAB



Average reward fell down in hard constraints by 13 points. Efficiency remains almost same.